

Chemotherapy for breast cancer

Chemotherapy uses anti-cancer (cytotoxic) drugs to destroy cancer cells.

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What is chemotherapy?

Chemotherapy uses anti-cancer (cytotoxic) drugs to destroy cancer cells. Cytotoxic means toxic to cells. Chemotherapy drugs disrupt the way cancer cells grow and divide, but they also affect normal cells.

When you have chemotherapy

You can have chemotherapy for breast cancer before or after surgery. Your cancer doctor or nurse will talk to you about the benefits of chemotherapy in your situation. They will also explain the likely side effects.

Before surgery

You may be offered chemotherapy before surgery if the cancer is large or growing quickly. This is called neo-adjuvant treatment. It may also be recommended if you have:

- [inflammatory breast cancer](#)
- [HER2 positive breast cancer](#)
- [triple negative breast cancer](#)
- breast cancer that has spread to the lymph nodes.

Neo-adjuvant chemotherapy aims to:

- shrink the size of the cancer
- reduce the risk of the cancer coming back
- avoid having to remove the whole breast (a mastectomy).

After surgery

Your cancer doctor may recommend you have chemotherapy after surgery. This is called adjuvant chemotherapy. It aims to reduce the risk of breast cancer coming back.

Your cancer doctor or breast cancer nurse will explain the benefits and likely side effects to you.

You are usually offered adjuvant chemotherapy if the cancer:

- has spread to the lymph nodes
- is large
- is [high grade](#)
- is [HER2 positive](#)
- is [triple negative](#).

You may also be offered it if you had a [tumour-profiling test](#) and the score was high.

How chemotherapy is given

You usually have chemotherapy in a chemotherapy day unit. You usually have it into a vein (intravenously). But some people are given chemotherapy tablets.

If you have chemotherapy intravenously, you have it by injection or as a drip (infusion). You may have:

- a [cannula](#) – a short, thin tube the nurse puts into a vein in your arm or hand
- a [PICC line](#) – a fine tube that is put into a vein in your arm and goes up into a vein in your chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on your chest or arm and goes into a vein in your chest.

You usually have chemotherapy as a few sessions of treatment. Each session takes a few hours. After the session, you have a rest period. The session and rest period together are called a cycle.

The length of a cycle depends on the chemotherapy drugs you are having. Most cycles are 1 to 3 weeks. Your cancer doctor or nurse will tell you how many cycles you need.

The drugs used

Different chemotherapy drugs affect cancer cells in different ways. This is why a combination of drugs is often used. For early breast cancer, the combination usually includes drugs called:

- anthracyclines – such as [epirubicin](#) and [doxorubicin](#)
- taxanes – such as [docetaxel](#), [paclitaxel](#), and [nab-paclitaxel](#).

These drugs are commonly used in combination with other chemotherapy drugs, including [cyclophosphamide](#) and [carboplatin](#).

If you are having both an anthracycline and a taxane, they will usually be given one after another. Your cancer doctor or nurse will explain the order you have them.

Some commonly used combinations are:

- [AC](#) – doxorubicin (Adriamycin®) and cyclophosphamide
- [EC](#) – epirubicin and cyclophosphamide
- [TC](#) – docetaxel (Taxotere®) and cyclophosphamide
- carboplatin with paclitaxel or docetaxel
- EC-T – epirubicin and cyclophosphamide, followed by docetaxel
- EC-P – epirubicin and cyclophosphamide, followed by paclitaxel.

Your cancer doctor may offer you a choice of chemotherapy drugs. They will give you information to help you decide.

If you have HER2 positive breast cancer, you will have targeted therapy drugs called [trastuzumab](#) or pertuzumab. Or you may have [both](#).

After surgery, you may continue having the drugs you had before surgery. Or you may change to [trastuzumab emtansine](#).

These may be given if there is a higher risk of the cancer coming back, to reduce the risk. We have more information about [targeted therapy for breast cancer](#).

Side effects of chemotherapy for breast cancer

Chemotherapy causes some [side effects](#). But these are usually well controlled with medicines. Most side effects usually go away after treatment has finished. Not all drugs cause the same side effects. Your cancer doctor, breast cancer nurse or pharmacist will explain what to expect from the treatment.

We have more information about the side effects of each [different drug](#).

Getting support

Macmillan is here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line for free on [0808 808 00 00](#).
- [Chat to our specialists online](#).
- Visit our [chemotherapy forum](#) to talk with people who have had chemotherapy, share your experience, and ask questions.

About our information

References



Below is a sample of the sources used in our breast cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

ESMO. Early breast cancer clinical practice guidelines for diagnosis, treatment and follow-up. 2019, Vol 30, pp1192–1220. Available from: <https://www.esmo.org/guidelines/guidelines-by-topic/breast-cancer/early-breast-cancer> [accessed 2023].

National Institute for Health and Care Excellence (NICE). Early and locally advanced breast cancer: diagnosis and management. 2018. Updated 2023. Available from: <https://www.nice.org.uk/guidance/ng101> [accessed 2023].

Reviewers

This information has been written, revised and edited by Macmillan Cancer Support’s Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by [Dr Rebecca Roylance](#), Consultant Medical Oncologist and [Professor Mike Dixon](#), Professor of Surgery and Consultant Breast Surgeon.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF’s 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 October 2023 | Next review: 01 October 2026

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)